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Language Model Fine-Tuning on Scaled Survey Data for Predicting Distributions of Public Opinions

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Abstract

Large language models (LLMs) present novel opportunities in public opinion research by predicting survey responses in advance during the early stages of survey design. Prior methods steer LLMs via descriptions of subpopulations as LLMs' input prompt, yet such prompt engineering approaches have struggled to faithfully predict the distribution of survey responses from human subjects. In this work, we propose directly fine-tuning LLMs to predict response distributions by leveraging unique structural characteristics of survey data. To enable fine-tuning, we curate SubPOP, a significantly scaled dataset of 3,362 questions and 70K subpopulation-response pairs from well-established public opinion surveys. We show that fine-tuning on SubPOP greatly improves the match between LLM predictions and human responses across various subpopulations, reducing the LLM-human gap by up to 46% compared to baselines, and achieves strong generalization to unseen surveys and subpopulations. Our findings highlight the potential of survey-based fine-tuning to improve opinion prediction for diverse, real-world subpopulations and therefore enable more efficient survey designs. Our code is available at https://github.com/JosephJeesungSuh/subpop.

1 Introduction

Surveys provide an essential tool for probing public opinions on societal issues, especially as opinions vary over time and across subpopulations. However, surveys are also costly, time-consuming, and require careful calibration to mitigate non-response and sampling biases (Choi and Pak, 2004; Bethlehem, 2010). Recent work suggests that large language models (LLMs) can assist public opinion studies by predicting survey responses across different subpopulations, explored in both social science (Argyle et al., 2023; Bail, 2024; Ashokkumar et al., 2024; Manning et al., 2024) and NLP (Santurkar et al., 2023; Chu et al., 2023; Moon et al., 2024; Hämäläinen et al., 2023; Chiang and Lee, 2023). Such capabilities could substantially enhance the survey development process, not as a replacement for human participants but as a tool for researchers to conduct pilot testing, identify subpopulations to over-sample, and test analysis pipelines prior to conducting the full survey (Rothschild et al., 2024).

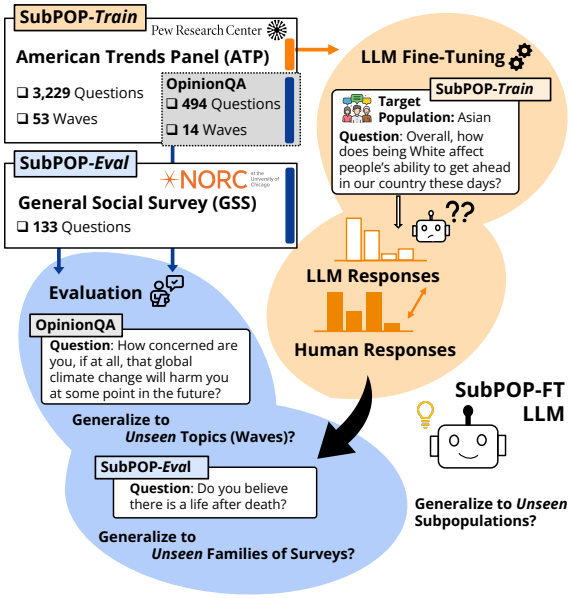

Figure 1: Illustration of our method and SubPOP. We collect survey data from two survey families—ATP from Pew Research (Pew Research Center, 2018) (forming SubPOP-Train) and GSS from NORC (Davern et al., 2024) (forming SubPOP-Eval). LLMs are fine-tuned on SubPOP-Train and evaluated on both OpinionQA (Santurkar et al., 2023) and SubPOP-Eval to assess generalization of distributional opinion prediction across unseen survey topics, survey families, and subpopulations.

Prior work in steering language models, *i.e.* conditioning models to reflect the opinions of a specific subpopulation, has primarily investigated different prompt engineering techniques (Santurkar et al., 2023; Moon et al., 2024; Park et al., 2024a).

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However, prompting alone has shown limited success in generating completions that accurately reflect the distributions of survey responses collected from human subjects. Off-the-shelf LLMs (Achiam et al., 2023; Dubey et al., 2024; Jiang et al., 2023) have shown to mirror the opinions of certain US subpopulations such as the wealthy and educated (Santurkar et al., 2023; Gallegos et al., 2024; Deshpande et al., 2023; Kim and Lee, 2023), while generating stereotypical or biased predictions of underrepresented groups (Cheng et al., 2023b,a; Wang et al., 2024). Furthermore, these models often fail to capture the variation of human opinions within a subpopulation (Kapania et al., 2024; Park et al., 2024b). While fine-tuning presents opportunities to address these limitations (Chu et al., 2023; He et al., 2024), existing methods fail to train models that accurately predict opinion distributions across diverse survey question topics and subpopulations.

The present work. Here, we propose directly fine-tuning LLMs on large-scale, high-quality survey data, consisting of questions about diverse topics and responses from each subpopulation, defined by demographic, socioeconomic, and ideological traits. By casting pairs of (subpopulation, survey question) as input prompts, we train the LLM to align its response distribution against that of human subjects in a supervised manner. We posit that survey data is particularly well-suited for fine-tuning LLMs since: (1) We can train the model with clear **subpopulation-response pairs** that explicitly link group identities and expressed opinions, which is rare in LLMs' pre-training corpora, (2) Large-scale opinion polls are carefully designed and calibrated (*e.g.* using post-stratification) to estimate **representative** human responses, in contrast with LLMs' pre-training data where certain populations are over- or underrepresented, (3) Our training objective explicitly aligns model predictions with response **distributions** from each subpopulation, enabling LLMs to capture variance within human subpopulations.

Training on public opinion survey data has remained under-explored due to the limited availability of structured survey datasets. To this end, we curate and release **SubPOP** (**Sub**population-level **P**ublic **O**pinion **P**rediction), a dataset of 70K subpopulation-response distribution pairs ($6.5\times$ larger compared to previous datasets). We show that fine-tuning LLMs on SubPOP significantly improves the distributional match between LLM generated and human responses, and improvements are consistent across subpopulations of varying sizes. Additionally, the improvement generalizes to *unseen* subpopulations, survey waves (topics), and survey families, *i.e.* surveys administered by different institutions. Such broad generalization is particularly critical for real-world public opinions research, where practitioners are most in need of synthetic data for survey questions or subpopulations (or both) that they have not tested before.

Our contributions are summarized as follows:

- We show that training LLMs on response distributions from survey data significantly improves their ability to predict the opinions of subpopulations, reducing the Wasserstein distance between LLM and human distributions by 32-46% compared to top-performing baselines. (Section 4.2)
- We show that the performance of the fine-tuned LLMs strongly generalizes to out-of-distribution data, including unseen subpopulations, new survey waves, and different survey families. (Section 4.2 and Section 4.3)
- We release SubPOP, a curated and pre-processed dataset of public opinion survey results that is $6.5\times$ larger than existing datasets, enabling fine-tuning at scale.

2 Related Work

Public opinion datasets. Several research institutions conduct large-scale public opinion polls and release data from those surveys. Important examples include Pew Research Center's American Trends Panel (ATP), which consists of multiple waves of cross-sectional surveys on different topics (Pew Research Center, 2018), and General Social Survey (GSS) from the NORC at the University of Chicago (Davern et al., 2024). Existing datasets have curated such data for evaluating LLM-based opinion predictions, including OpinionQA (Santurkar et al., 2023), a subset of ATP survey waves containing about 500 questions on contentious social topics. While OpinionQA is widely used in prior work (He et al., 2024; Zhao et al., 2023; Li et al., 2023, 2024), we find its total number of questions limited in scale for fine-tuning LLMs and instead use this dataset for evaluation. We further collect an extended set of survey data from ATP waves not included in OpinionQA, as well as from GSS to curate SubPOP. Other datasets, such as GlobalOpinionQA (Durmus et al., 2023)—derived from the World Values Survey (WVS)

(World Values Survey, 2022) and the Pew Global Attitudes Survey (Pew Research Center, 2024)—and the PRISM dataset (Kirk et al., 2024) investigates how language models align with opinions from populations across the globe and different cultures.

Predicting human opinions with LLMs. Prior work has explored various prompt engineering approaches for steering LLM responses: earlier work use rule-based prompts that incorporate demographic profiles of individuals or populations, or few-shot examples of survey question-response (Hwang et al., 2023; Simmons, 2022; Santurkar et al., 2023; Dominguez-Olmedo et al., 2023). Recent work explore prompting LLMs with open-ended text, including interview transcripts (Park et al., 2024a), personal narratives (Moon et al., 2024), or LLM-refined prompts (Kim and Yang, 2024; Sun et al., 2024). Our fine-tuning approach is complementary to prompt engineering methods: while prompt engineering seeks to optimize what information is provided to the LLM (while the model is frozen), fine-tuning seeks to optimize how the model utilizes the provided information (while the prompt is frozen). In this work, we demonstrate that our fine-tuned models exhibit significant improvements in matching the response distributions of humans without requiring elaborate prompt engineering methods.

Other work (Chu et al., 2023; He et al., 2024; Feng et al., 2024) fine-tune language models on text corpora from specific communities (*e.g.*, Reddit) to infer the most popular response or response distribution for a given survey question. While this approach benefits from large-scale and continuously updated text corpora, it struggles with disproportionate representation online and lacks comprehensive coverage of diverse subpopulations. A few works have explored directly fine-tuning on public opinion survey data, but in different problem settings from ours. Li et al. (2023) apply collaborative filtering to individual-level responses to learn embeddings for individuals, and Zhao et al. (2023) develop a meta-learning framework to predict the opinions of new groups given a small number of in-context examples for that group. In contrast, our approach does not require individual-level responses and can generalize to unseen groups and survey questions without *any* responses.

A recent work (Li et al., 2024) and a work concurrent to ours (Cao et al., 2025) also explored fine-tuning LLMs on the World Values Survey (WVS) to align the LLM's opinion response with a culture or entire country populations. In comparison, our work focuses on US surveys, testing whether LLMs can align with finer-grained subpopulations within one country and whether LLMs fine-tuned on one US-representative survey can generalize to another. However, we note that our proposed method for fine-tuning language models applies to any survey dataset with distributional information about subpopulation responses.

Pluralistic alignment of LLMs. Recent literature on pluralistic and distributional alignment target a similar yet different problem in fine-tuning LLMs (Chakraborty et al., 2024; Melnyk et al., 2024; Poddar et al., 2024; Siththaranjan et al., 2023; Yao et al., 2024; Sorensen et al., 2024; Lake et al., 2024; Chen et al., 2024; Jiang et al., 2024). While this line of work shares a similar goal as ours in training models to reflect on opinions (and preferences) of diverse subpopulations, most work differ from ours in that they operate in the context of training against *pair-wise* preference orderings between alternative language model completions, extending the Bradley-Terry-Luce model (Rajkumar and Agarwal, 2014; Ouyang et al., 2022; Rafailov et al., 2024) or investigating alternative models to account for diverging preference orderings across populations. In contrast, our work trains the model to directly predict the opinion distributions of human subpopulations, where accurately matching distributions across a large variety of subpopulations is of paramount interest. Our work additionally focuses on the particular context of estimating human opinions about societal issues—the objective of public opinion research—which enables relatively straightforward supervised training on openly available, structured survey data as presented by `SubPOP`.

3 Methods

3.1 Fine-tuning LLMs on Human Response Distributions

Our goal is to fine-tune an LLM to predict the distribution of responses for a multiple-choice question, conditioned on descriptions of a human subpopulation we want to simulate, typically a specific demographic, socioeconomic, or ideological subgroup. Consider the example in Figure 2: the question asks, "What do you think the chances are these days that a woman won't get a job or promotion while an equally or less qualified man gets one instead?" The